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Broad-Based Asset Management Programs¹

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Yale Program on Financial Stability Survey

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Abstract

Dealing with high levels of nonperforming assets (NPAs) on bank balance sheets is one of the most challenging aspects of financial crisis management. High levels of NPAs can interfere with both bank profitability and general economic growth by increasing uncertainty about bank solvency and therefore funding costs, tying up resources and attention, and inhibiting new lending. One potential solution to the NPA problem is a centralized, government-driven effort to remove these assets from troubled institutions and then manage and sell them. Though such broad-based asset management (BBAM) programs existed even earlier in history, they appear to have become more common beginning in the 1980s and 1990s with the shift toward market-based financial systems in Africa, South America, and the former Soviet bloc, as well as with the advent of the Asian Financial Crisis. They were also a feature of the response to the Global Financial Crisis. While BBAM programs have been widely used, whether or not such a program makes sense to address a given situation is highly context-specific, and special attention must be paid to the incentives of those involved. Important considerations include the political and legal context in which the program will operate, the nature and extent of the NPAs to be managed, and the availability of the necessary expertise. There is also evidence to suggest that BBAM programs are ineffective in isolation and must be coupled with recapitalization.

Keywords: asset management companies, nonperforming loans, nonperforming assets

¹ This survey analyzes case studies that are part of the Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering broad-based asset management programs. Cases are available from the Journal of Financial Crises at <https://elischolar.library.yale.edu/journal-of-financial-crises/>.

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Introductory Note

This survey is an analysis of important considerations for policymakers seeking to establish a broad-based asset management (BBAM) program. It is based on insights derived from case studies of 29 specific BBAM programs the Yale Program on Financial Stability has completed and from the existing literature on the topic. While this survey can help inform a decision about whether or not to establish a BBAM program, it is not intended to be definitive on this question. Instead, the survey is intended primarily to assist policymakers who have already made that decision in designing the most effective program possible.

In analyzing the programs that are the focus of this survey, we used a color-coded system to highlight certain particularly noteworthy design features. Our color-coding system is as follows:

Color	Meaning
BLUE – INTERESTING	A design feature that is interesting and that policymakers may want to consider. Typically, this determination is based on the observation that the design feature involves a unique and potentially promising way of addressing a challenge common to this type of program that may not be obvious. Less commonly, there will be empirical evidence or a widely held consensus that the design feature was effective in this context, in which case we will describe that evidence or consensus.
YELLOW – CAUTION INDICATED	A design feature that policymakers should exercise caution in considering. Typically, this determination is based on the observation that the designers of the feature later made significant changes to the feature with the intention of improving the functioning of the program. Less commonly, there will be empirical evidence or a widely held consensus that the design feature was ineffective in this context, in which case we will describe that evidence or consensus.
FOOTNOTE IN ITALICS	Where the reason that a given design feature has been highlighted is not apparent from the text, it is accompanied by a footnote that explains why we chose to highlight it as we did. Where necessary,

	footnotes will also be used to identify any caveats or additional considerations that should be kept in mind when thinking about the feature. These footnotes are italicized to identify them for the reader.
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This highlighting is not intended to be dispositive. The fact that a design feature is not highlighted or is highlighted yellow does not mean that it should not be considered or that it will never be effective under any circumstances. Similarly, the fact that a design feature is not highlighted or is highlighted blue does not mean that it should always be considered or will be effective under all circumstances. The highlighting is our subjective attempt to guide readers towards certain design features that (1) may not be obvious but are worth considering or (2) require caution in considering. Readers must always consider these features while keeping in mind the contents of the survey as a whole and their own particular context and objectives.

I. Overview

The question of what to do with the nonperforming assets (NPAs)⁶ that often plague bank balance sheets during banking crises may be “one of the most critical and complex aspects of financial sector crisis management” (Ingves, Seelig, and He 2004). High levels of NPAs increase uncertainty about bank solvency and bank funding costs, tie up resources, and constrain new lending (Fell et al. 2017). They can also interfere with the transmission of monetary policy (Fell et al. 2017). Not surprisingly then, evidence has shown that elevated NPA percentages slow economic growth (Balgova, Nies, and Plekhanov 2016).

Faced with the need to ensure the effective management and disposition of NPAs, policymakers have frequently determined that a centralized, government-driven approach to dealing with such assets is preferable to any decentralized alternatives. A key question is whether the centralized program will be able to achieve the improved outcomes necessary to justify the cost and effort (which are often quite significant). With that in mind, governments typically choose a centralized approach when (1) their troubled-asset problem is systemic, rather than limited to one or two banks; (2) their troubled assets are focused on sectors—particularly commercial real estate and large corporate loans—in which a centralized program could have a competitive advantage due to bargaining power, economies of scale, or the homogeneity of the assets; and (3) they have or believe they will be able to acquire the necessary expertise to manage and dispose of those assets.

⁶ This survey uses the term “nonperforming assets” instead of “nonperforming loans” to reflect the fact that while the assets acquired by BBAM programs were typically loans, other types of assets were sometimes involved. See Eligible Assets under Key Design Decisions for additional discussion.

Given the various difficulties presented by high levels of NPAs, governments' objectives in pursuing a centralized approach to removing NPAs can vary. The specific objectives being sought by a particular effort will shape how it should be designed. Key objectives that often underlie governments' removal of NPAs from bank balance sheets include:

1. Reducing uncertainty about the quality of banks' balance sheets and protecting them from further losses
2. Allowing banks to focus on their core business rather than the management of NPAs
3. Cleaning up particular sectors of the economy that have resulted in NPAs
4. Minimizing the ultimate cost to taxpayers of the NPAs
5. Managing the political and communication challenges posed by the NPAs and efforts to address them.

While there are examples of governments establishing vehicles to remove problematic assets from institutions' balance sheets even earlier in history,⁷ the practice appears to have become more common beginning in the 1980s and 1990s. During this timeframe, countries in Africa, South America, and the former Soviet bloc that were transitioning to more market-based banking systems had to confront large stocks of legacy NPAs stemming from prior periods of government-directed lending, and they founded asset management programs in response. The Asian Financial Crisis in 1997 also gave rise to several such programs. Similarly, these programs were often a feature of countries' responses to the Global Financial Crisis.

This survey is limited to the examination of broad-based asset management (BBAM) programs and therefore excludes interventions such as the 2008 Maiden Lane vehicles in the United States intended to address the problematic assets of a specific institution.⁸ Table 1 shows a list of the BBAM programs studied for the purposes of this survey. This list includes programs that acquired assets solely from banks that remained in operation (open-bank-only programs), programs that acquired assets solely from failed banks being resolved (closed-bank-only programs), and programs that acquired assets from both types of banks. The considerations associated with an open-bank-only program are not identical to those associated with a closed-bank-only program, particularly in terms of acquisition. However, these different types of programs have enough in common to warrant being examined together.

⁷ For a data set of crisis interventions (including broad-based asset management programs) dating back to the 14th century, see Metrick and Schmelzing (working paper).

⁸ Included in the BBAM programs studied in this survey are programs such as Sweden's Securum/Retriva and China's 1999 AMCs, in which policymakers initially established vehicles to address the problematic assets of particular institutions on a one-to-one basis, before such vehicles were ultimately consolidated or otherwise assumed responsibility for a broader array of financial institutions.

Table 1: List of BBAM Programs Studied

Country	Program	Year Launched	Open or Closed Bank
Spain	Deposit Guarantee Fund (DGF)	1980	Closed
Uruguay	Central Bank of Uruguay's (CBU) Nonperforming Portfolio Purchase Scheme (NPPS)	1982	Closed
Senegal	Société Nationale de Recouvrement (SNR)	1989	Closed
United States	Resolution Trust Corporation (RTC)	1989	Closed
Ghana	Nonperforming Asset Recovery Trust (NPART)	1990	Both
Burkina Faso	Bureau de Recouvrement des créances du Burkina (BRCB)	1991	Both
Czechoslovakia	Consolidation Bank (KOB)	1991	Open
Tanzania	Loans and Advances Realization Trust (LART)	1991	Open
Hungary	Loan Consolidation Programs (LCPs)	1992	Open
Sweden	Securum/Retriva	1992	Both
Finland	Arsenal	1993	Closed
Kazakhstan	Rehabilitation Bank (RB)	1995	Both
Mexico	Fondo Bancario de Protección al Ahorro (FOBAPROA)	1995	Open
Mongolia	Mongolian Asset Realization Agency (MARA)	1996	Closed
Kyrgyz Republic	DEBRA Debt Resolution Unit (DRU)	1996	Both
Jamaica	Financial Sector Adjustment Company (FINSAC)	1997	Both
South Korea	Korea Asset Management Corporation (KAMCO)	1997	Open
Indonesia	Indonesian Bank Restructuring Agency (IBRA)	1998	Both
Japan	Resolution and Collection Corporation (RCC)	1998	Both
Malaysia	Pengurusan Danaharta Nasional Berhad (Danaharta)	1998	Open
China	Four AMCs	1999	Open
Colombia	Central de Inversiones (CISA)	2000	Both
Thailand	Thai Asset Management Company (TAMC)	2001	Open
Ireland	National Asset Management Agency (NAMA)	2009	Open
Nigeria	Asset Management Corporation of Nigeria (AMCON)	2010	Open
United Kingdom	United Kingdom Asset Resolution (UKAR)	2010	Closed
Slovenia	Bank Asset Management Company (BAMC/DUTB)	2012	Open
Spain	La Sociedad de Gestión de Activos Procedentes de la Reestructuración Bancaria (SAREB)	2012	Both
Hungary	Magyar Reorganizációs és Követeléskezelő (MARK)	2014	Open

Source: Author analysis.

As the launch dates in Table 1 suggest, there were distinct periods of BBAM program usage (Asian Financial Crisis, Global Financial Crisis, etc.). Programs developed in later periods of usage could and often did incorporate lessons learned from previous efforts. Policymakers have often cited Sweden's Securum/Retriva and the United States' Resolution Trust Corporation (RTC) as models for their own efforts.

Given the critical nature of NPA problems, the fact that BBAM programs were not even more widespread merits discussion. Whether or not it makes sense to address a given situation with a BBAM program is highly context specific. Certainly there are considerations that could weigh in favor of alternate approaches like keeping NPAs with the banks who originated them for resolution or using asset management companies (AMCs) specific to each institution. Such decentralized AMCs were more common than the centralized AMCs

characteristic of most BBAM programs in the Global Financial Crisis (Dobler, Moretti, and Piris 2020). Table 2 summarizes some of the main factors to be considered in determining whether to employ a BBAM program over such alternatives.

Table 2: Factors to Weigh in Considering a BBAM Program

Factor	BBAM Generally Indicated	BBAM Generally Not Indicated
Political culture	Tradition of institutional independence and public accountability	Tradition of political interference and an absence of accountability
Legal system	Bankruptcy, insolvency, and foreclosure laws strong enough (or capable of being made strong enough) for creditors to enforce rights	Bankruptcy, insolvency, and foreclosure laws not capable of being made strong enough for creditors to efficiently enforce rights
Nature of NPA problem	Commercial real estate	Loans to state-owned enterprises or other politically connected debtors
	Large corporate exposures	Residential mortgages
		Small and medium-sized enterprise loans
		Consumer loans
Extent of NPA problem	Large enough to justify the effort and expense of a centralized body	Too small to justify the effort and expense of a centralized body
	Not so large as to make political interference unavoidable and/or to limit the potential buyers of the assets	Too large to avoid political interference and/or to ensure potential buyers of the assets
Expertise	Not enough potential expertise to make available on a decentralized basis	A desire/the ability to develop workout expertise at the bank level

Sources: Baudino and Yun 2017; Cas and Peresa 2016; Cerruti and Neyens 2016; EC 2018; Fell et al. 2017; Ingves, Seelig, and He 2004; Klingebiel 2000; Rose 2005.

In considering these factors, careful attention must be paid to the incentives of all parties that would be involved in a BBAM program and how they compare with the incentives that would exist absent the program—the incentives of the banks, bank employees, program employees, and debtors. For example, the incentives of BBAM program employees (who, among other considerations, would typically be out of a job following the disposal of the acquired NPAs) are different from those of bank employees should NPAs remain with banks.

II. Key Design Decisions

Part of a Package

As outlined in the Background section, high levels of NPAs can have significant negative consequences for financial institutions. It is thus not surprising that the BBAM programs studied were typically not introduced in isolation, but rather alongside other measures intended to address some of the consequences that can result from NPAs including the loss of liquidity and the erosion of capital. Interventions such as emergency lending facilities, asset guarantees, liability guarantees, corporate restructuring initiatives, recapitalizations, banking sector restructuring, and bank regulatory reform are common counterparts to BBAM initiatives. Table 3 below shows which of the programs studied were combined with these other interventions.

Table 3: Program Packages

BBAM Program	Recapitalization	Bank Resolution & Restructuring	Corporate Restructuring	Regulatory Reform	Emergency Lending	Guarantee (Asset or Liability)
Burkina Faso: BRCB	X	X		X		
China: Four AMCs	X			X		
Colombia: CISA	X	X			X	
Czechoslovakia: KOB	X					
Finland: Arsenal	X	X				X – Both
Ghana: NPART	X	X	X	X		
Hungary: LCPs	X		X	X		
Hungary: MARK						
Indonesia: IBRA	X		X			X – Liability
Ireland: NAMA	X					
Jamaica: FINSAC	X	X		X		
Japan: RCC	X	X	X	X		
Kazakhstan: RB			X	X		
Kyrgyz Republic: DEBRA DRU	X	X		X		
Malaysia: Danaharta	X		X			
Mexico: FOBAPROA	X					
Mongolia: MARA	X	X				
Nigeria: AMCON	X					
Senegal: SNR		X		X		
Slovenia: BAMC/DUTB	X			X		
South Korea: KAMCO	X		X	X		
Spain: DGF	X	X		X		X – Liability
Spain: SAREB	X	X		X		
Sweden: Securum/ Retriva	X	X				X – Liability
Tanzania: LART	X					
Thailand: TAMC						
United Kingdom: UKAR		X				
United States: RTC		X		X		
Uruguay: CBU		X				

Source: Author analysis.

The decision about what to include in the package follows directly from the diagnosis of the problem, particularly the extent and type of NPAs. A country with a relatively small NPA problem focused on commercial real estate might require only a voluntary BBAM program, tied to voluntary recapitalization. A country with a larger NPA problem focused on corporate loans might also need a major corporate restructuring initiative. A country whose banking troubles are focused on a small number of systemically important institutions might be better served by company-specific rescue plans and a BBAM program only for certain types of commonly held assets, which may lend themselves to economies of scale.

Of the potential measures that could join BBAM programs as part of a broader bank support package, recapitalization is often the single most important counterpart. Banks with high levels of NPAs, almost by definition, face capital shortfalls. The exact need for recapitalization is closely tied to a BBAM program's approach to pricing. As discussed in more detail in the section on Pricing, policymakers must decide whether to acquire NPAs at book value or for some lower amount, such as real economic value or market price. The former approach protects transferring institutions from losses on the sale of NPAs and shifts those losses to the BBAM program. This acts as a recapitalization of the transferring institutions and can eliminate the need for a separate capital injection program (but can also have significant drawbacks as will be discussed in the Pricing section).

Acquiring assets for some lower amount, by contrast, imposes losses on the transferring institutions. Depending on the size of those losses, an institution's capital may become so eroded as to require recapitalization. The failure to pair a BBAM program with a recapitalization measure under these circumstances would likely prevent policymakers from achieving their desired outcomes. As will be discussed in more detail in the Evaluation section, an analysis of asset segregation efforts in Europe from 2000 to 2016 suggests that the removal of NPAs without recapitalization does not improve loan growth or reduce future NPA levels. The study finds that the reverse is also true—that recapitalization without removal of NPAs is ineffective (Brei et al. 2020). The BBAM programs studied here likewise provide evidence of a need to combine NPA removal with recapitalization. Hungary's Loan Consolidation Program (LCPs), launched in 1992, failed to sufficiently reduce NPA levels by itself, resulting in the introduction of a recapitalization measure in 1993–1994. Kazakhstan's Rehabilitation Bank similarly began acquiring NPAs without capital injections. The country's largest banks remained under significant financial stress and ultimately had to be rescued.

The positive interplay between BBAM programs and recapitalization measures may mean that eligibility for one should be contingent on participation in the other. Several of the programs studied adopted this approach. For Mexico's Fondo Bancario de Protección al Ahorro (FOBRAPROA), recapitalization was not only required for participation in the BBAM program, but it also determined the amount of NPAs to be acquired from a given institution—approximately two times the amount of new capital it received.⁹

For efforts combining BBAM programs with other measures such as recapitalizations, the next important consideration is whether the initiatives should be combined under a single entity/umbrella or split up among different entities/umbrellas, each with a singular focus. The latter approach appears to have been more common among the programs studied. Malaysia, for example, established Pengurusan Danaharta Nasional Berhad (Danaharta) to remove NPAs from the system and Danamodal Nasional Berhad to recapitalize banks. As will be discussed in Mandates below, a combined approach may give rise to concerns about conflicting mandates and a loss of focus.

⁹ In the case of FOBRAPROA, most of this capital was provided by private sources with government encouragement rather than coming in the form of a direct recapitalization by the state.

Legal Authority

The removal of troubled assets from the balance sheets of financial institutions is not typically something that is within the ongoing mandates of central banks or other governmental bodies. Thus, the majority of the programs studied required new laws in order to be established. One unique exception to this general approach was Hungary's Magyar Reorganizációs és Követeléskezelő (MARK), which the Hungarian central bank created pursuant to its macroprudential policy mandate.¹⁰ In Thailand, policymakers made use of a royal decree to expedite establishment of the Thai Asset Management Company (TAMC). Uruguay's central bank appears to have relied on unclear legal authority to launch the Nonperforming Portfolio Purchase Scheme (NPPS) initially before it was ultimately ratified through new legislation. This resulted in several investigations by Parliament, the comptroller, and the court system. The typical need to pass new legislation authorizing BBAM programs and the added time and difficulty this can involve especially in crisis conditions may mean that it is worth considering proactively establishing the legal basis for programs to be created under certain circumstances. This could be done before the next crisis hits.

An additional consideration for BBAM programs established in the European Union (EU) has been the need to comply with European Commission (EC) rules on State aid.¹¹ These rules significantly influenced program design, especially with respect to the price to be paid for NPAs, as discussed in greater detail in the Pricing section.

Special Powers

As noted in the Overview, the legal context in which a BBAM program operates significantly influences its likelihood for success. An evaluation of the robustness of its legal framework is an important step for any country contemplating a BBAM, with particular attention to whether insolvency, bankruptcy, and foreclosure laws are strong enough to allow creditors to efficiently pursue recoveries. Sometimes this evaluation will result in the conclusion that the existing framework is sufficient, as was the case with Sweden's Securum/Retriva, where policymakers explicitly determined that special powers were unnecessary. However, oftentimes countries introducing BBAM programs lacked effective legal frameworks for dealing with NPAs. This is often the case for emerging markets and developing economies. It appears to have been particularly true for countries whose BBAM programs were introduced as part of making the shift toward market-based financial systems. One way in which policymakers sought to overcome deficiencies in their existing frameworks was to equip BBAM programs with special powers. BBAM programs can generally be granted enhanced legal authority more quickly than a country can completely rework its existing legal

¹⁰ The use of an existing authority such as a macroprudential policy mandate to establish a BBAM program may allow for more rapid introduction of a program during a crisis in which speed is often of the essence, but should be weighed against other potential considerations such as the need for the political legitimacy that can result from specific legislative authorization of a program.

¹¹ The legal requirements for BBAM programs adopted in the EU have continued to evolve since the time of the programs studied in this survey with the introduction of the Bank Recovery and Resolution Directive and the Single Resolution Mechanism Regulation. For a complete discussion of these requirements, see EC 2018.

framework, which may be one reason to consider adopting a program. The need to provide special powers might also weigh in favor of a BBAM program that is fully publicly owned.

Table 4 provides an overview of the various special powers included in the programs studied. As Table 4 illustrates, special powers were typically aimed at addressing deficiencies associated with inefficient or nonexistent legal processes for dealing with delinquent debtors. Often this became most important in the context of corporate debt. The problem of politically connected debtors was also a special problem in many cases.

Sometimes deficiencies requiring special powers did not reveal themselves until after the BBAM programs were already underway, making program amendments necessary. Burkina Faso's Bureau de Recouvrement des créances du Burkina (BRCB) acquired enhanced powers to recover debt two years into its operations. Despite an initial grant of some special powers, Nigeria's Asset Management Corporation of Nigeria (AMCON) required enhancements to its initial powers as time went on, including the ability to seize any debtor assets and the right to access debtors' financial information.

Table 4: Summary of Special Powers

Authority area	Description	BBAM Programs
Courts	Special tribunal/court	Ghana: NPART Tanzania: LART
	Expedited court access	Mongolia: MARA
	Bypassed court procedures	Malaysia: Danaharta Thailand: TAMC
	Staff immunity from criminal liability	Thailand: TAMC
	Decisions could not be challenged in court / were considered final	Ghana: NPART Tanzania: LART
	Courts prohibited from granting an injunction against the AMC	Ireland: NAMA Malaysia: Danaharta
	Ability to join legal proceedings on behalf of the financial institution	Ireland: NAMA Malaysia: Danaharta
	Summary judgments against debtors	Senegal: SNR
Recovery and restructuring	Enhanced powers to seize assets, restructure borrowers, or restructure debt	Burkina Faso: BRCB Colombia: CISA Indonesia: IBRA Ireland: NAMA Japan: RCC Kazakhstan: RB Malaysia: Danaharta Nigeria: AMCON (expanded powers in 2019 to seize any assets from a debtor) Senegal: SNR Slovenia: BAMC/DUTB Thailand: TAMC
	Publication of delinquent borrowers	Burkina Faso: BRCB Nigeria: AMCON
	Special investigative authority	Japan: RCC
	Access to debtor data or information	Ireland: NAMA Nigeria: AMCON
	Restitution from bank leaders and shareholders	Kyrgyz Republic: DEBRA
	Enacting new regulations	Ghana: NPART
Other legal authority	Tax exemptions	Ghana: NPART Senegal: SNR
None	No special powers granted	China: Four AMCS Czechoslovakia: KOB Finland-Arsenal Hungary: LCPs Hungary: MARK Jamaica: FINSAC South Korea: KAMCO Spain: SAREB Spain: DGF Sweden: Securum/Retriva United Kingdom: UKAR United States: RTC Uruguay: CBU

Source: Author analysis.

The efficacy of these special powers appears to be mixed. The World Bank has credited Tanzania's special tribunal and its simplified procedures and exclusive jurisdiction with having contributed to the success of the Loan and Advances Realization Trust (LART) (World Bank 2001). This success occurred despite the fact that LART dealt primarily with loans to state-controlled businesses, which, as discussed in Eligible Assets, can be a challenging asset class to address via a BBAM program. In Malaysia, Danaharta's special legal authority enabled it to reduce the time needed to restructure loans from 14 to 16 months to two to three months. However, special powers can only work if people are willing and able to use them, which was not always the case. Indonesia's Indonesian Bank Restructuring Agency (IBRA) possessed powerful enforcement mechanisms but was (1) generally unwilling to go after politically connected debtors and (2) often prevented from utilizing such mechanisms by Indonesian courts opposed to their use. Mongolia established a special tribunal as part of the Mongolian Asset Realization Agency (MARA), but its usefulness was limited by an inexperienced judiciary. Evaluations of Ghana's Nonperforming Asset Recovery Trust (NPART) are split on the effectiveness of its special tribunal, with some arguing that it helped mitigate the negative impact of a weak legal framework (World Bank 1997) and others arguing that it was too slow and debtor friendly (Klingebiel 2000). Additionally, special powers may not be enough to overcome underlying NPAs that are of especially poor quality. Burkina Faso's BRCB recovered only 10% of the value of the assets it acquired, notwithstanding what the International Monetary Fund (IMF) characterized as its powerful tools. Special powers can also be the source for political opposition. This was the case with Thailand's TAMC, whose special powers were criticized by opponents as being excessive. Perhaps to avoid such concerns, Malaysia's Danaharta had an oversight committee charged with making sure that special powers were not abused.

Mandate

The common mandate that all BBAM programs share is the removal of assets from the balance sheets of troubled financial institutions to incentivize new capital and allow banks to lend. In every program studied, the government was seeking to clean up banks' balance sheets or simply taking over the assets of banks already in liquidation. However, BBAM programs often differ in what they are expected to do with those assets once they take them over. Broadly speaking, policymakers generally task BBAM programs with either rapid disposal of acquired NPAs or the more active management of such NPAs via restructuring. Considerations guiding this choice can include the minimization of taxpayer losses, the maximization of NPA value, the avoidance of further market disruption via fire sales, and the creation of new financial products.

Governments also often give BBAM programs additional mandates. Mongolia's MARA, for example, sought to normalize the enforcement of contractual obligations as the country entered into its post-Soviet future. However, the inclusion of additional mandates beyond the main focus of a BBAM program creates the risk of conflicts or distractions. In the United States, Congress charged the RTC with maximizing the return on asset disposition efforts but also gave it additional mandates, including the promotion of affordable housing and the protection of local real estate markets. Some observers believe these conflicting mandates reduced RTC's effectiveness, given that, for example, reserving real estate assets for low-

income buyers could increase RTC's costs while concern for local real estate markets might delay sales (FDIC 1998).

Table 5 summarizes the various mandates for the programs studied.

Table 5: Mandates

BBAM Program	Rapid Disposal	Minimize Losses or Maximize Recovery	Debt Restructuring/ Management	Other Mandate
Burkina Faso: BRCB		X	X	
China: Four AMCs		X	X	
Colombia: CISA		X	X	
Czechoslovakia: KOB			X	
Finland: Arsenal		X		Investigating debtor irregularities
Ghana: NPART	X		X	
Hungary: LCPs			X	Ensuring survival of strategic companies
Hungary: MARK		X	X	
Indonesia: IBRA		X	X	
Ireland: NAMA		X		Restoring confidence in the banking sector; improving credit availability; socioeconomic development
Jamaica: FINSAC	X	X	X	
Japan: RCC		X	X	
Kazakhstan: RB			X	
Kyrgyz Republic: DEBRA DRU			X	
Malaysia: Danaharta		X		
Mexico: FOBAPROA		X		
Mongolia: MARA		X		Setting an example of financial discipline; using the legal system to pursue delinquent borrowers to normalize contractual obligations
Nigeria: AMCON		X		
Senegal: SNR				Banking sector reform
Slovenia: BAMC/DUTB		X	X	
South Korea: KAMCO		X	X	
Spain: DGF	X	X		
Spain: SAREB		X	X	
Sweden: Securum/Retriva		X	X	Stabilizing the financial and real estate markets; allowing good bank components to continue operations
Tanzania: LART		X	X	
Thailand: TAMC	X	X	X	
United Kingdom: UKAR		X		Promoting healthy competition; monitoring financial stability impact
United States: RTC		X		Minimizing impact on real estate markets; promoting affordable housing
Uruguay: CBU				Avoiding bankruptcy and intervention in failed banks

Source: Author analysis.

As noted in the Part of a Package section, one key consideration is whether a BBAM program will be combined under the same entity/umbrella with other interventions such as recapitalization. This too can result in conflicting mandates. Jamaica's Financial Sector Adjustment Company (FINSAC) was responsible for all aspects of the country's comprehensive effort to stabilize its banking system, including bank restructuring and recapitalization. Critics maintain that this complicated FINSAC's operations (Escobar et al. 1999). In Slovenia, policymakers initially planned to task the Bank Asset Management Company (BAMC) with both recapitalization and the acquisition of NPAs. The Ministry of Finance subsequently decided that the government would undertake recapitalization directly, allowing BAMC to concentrate on NPAs.

An additional mandate-related consideration in some cases was whether or not the BBAM program should be responsible for handling NPAs associated with particular, politically powerful debtors. One risk of having a BBAM program handle such NPAs is that the program could become a bailout for well-connected debtors or at least be perceived as such. As a distinct subcomponent of its overall operations, South Korea's Korea Asset Management Corporation (KAMCO) acquired the debt of the failed Daewoo Group, at the time one of the country's largest and most connected conglomerates. Critics have argued that KAMCO's purchase of Daewoo debt was politically motivated and contributed to the program's underperformance when the debt proved difficult to resolve (He 2004). In contrast, policymakers excluded the politically sensitive Rumasa conglomerate from the asset management activities of Spain's Deposit Guarantee Fund (DGF), arguing that including it would go beyond the scope of the Fund's mission. This decision has been widely praised by international observers (Sheng 1996).

Communication

In announcing the establishment of a BBAM program, policymakers face a number of potential risks related to how the program will be perceived. Acknowledging the existence of a significant NPA problem could worsen panic in the financial system. Rightly or wrongly, the program may be seen by some as a bailout for banks (as was the case with Ireland's National Asset Management Agency [NAMA]) or politically connected debtors (as was the case with Indonesia's IBRA). These risks and the ongoing need to maintain popular support highlight the importance of developing an effective communication strategy as part of a BBAM program.

Successful BBAM programs were typically transparent with the public about their goals and their progress in meeting those goals. Experts and international organizations have praised BBAM programs such as Sweden's Securum/Retriva, Malaysia's Danaharta, and South Korea's KAMCO for their transparency and the confidence this instilled in the public. Examples of programs that have been criticized for their lack of transparency include Indonesia's IBRA, Nigeria's AMCON, and Thailand's TAMC. Appropriate accounting and data standards can help improve transparency by ensuring that information is provided in a way that can be understood and trusted (EC 2018).

For some of the programs we studied, more specific information is available about their communications strategy. One key message appears to have been that the removal and management of NPAs was necessary for the appropriate functioning of the financial system and economy. Perhaps most notably, Thailand's TAMC featured very prominently in the 2001 election for prime minister, with the winning candidate campaigning on the message that TAMC was necessary to accelerate the country's economic recovery. Depositor protection was a key theme of messaging for the RTC in the United States and the Deposit Guarantee Fund in Spain. Given its focus on consumer assets, the United Kingdom Asset Resolution (UKAR) in the United Kingdom emphasized the consumer-friendly nature of its practices.

In Indonesia, IBRA's initial attempt to restrict communications about its activities resulted in continued negative market sentiment and considerable criticism of the program by detractors who mistakenly believed that IBRA was non-operational. It may also have interfered with the work of IBRA officials onsite with targeted banks.

Ownership Structure

BBAM programs take on NPAs in crisis conditions, often where no market for such assets exists and with the risk of significant further losses. Private participation in the ownership of BBAM programs can thus be difficult to secure. There can be additional, affirmative reasons for government ownership of BBAM programs. Public ownership of a BBAM program may make it more feasible to invest the program with special legal powers. Additionally, the absence of private interests may more easily allow for considerations beyond maximizing return (although as discussed in the Mandate section, drifting too far from a commercial purpose may hinder BBAM efficacy). Whether for some or all of these reasons, the vast majority of BBAM programs studied were publicly owned.

For the minority of programs that involved at least some private ownership, the reasons appear to have been largely related to legal and accounting considerations. Ireland's NAMA was 51% privately owned in response to then-current guidance from Eurostat¹² that asset management companies that were majority privately owned would be kept off the government's balance sheet. Spain's La Sociedad de Gestión de Activos Procedentes de la Reestructuración Bancaria (SAREB) had a 55% private ownership majority for the same reason. Slovenia's BAMC had hoped to adopt a similar approach, but the state-dominated nature of the Slovenian banking sector made private participation difficult to secure and BAMC ended up being fully publicly owned.¹³

¹² Eurostat is the statistical office of the European Union and "ensures the production of European statistics according to established rules and statistical principles."

¹³ *This highlighting is not intended as an assertion that private ownership of BBAM vehicles is a promising approach. Instead, it represents the view that if private ownership is pursued, it may be difficult to actually achieve in a banking system dominated by the state.*

Despite majority private participation, ownership can be structured so as to leave the state with most of the control. NAMA, for instance, was largely government controlled despite the 51% private ownership stake.¹⁴

The legal considerations surrounding private participation in the ownerships of BBAM programs in the European Union are continuing to evolve. It is important to note that notwithstanding Eurostat's guidance, credit rating agencies still considered BBAM program debt to be on government balance sheets, which calls into question the usefulness of this approach. In any event, the approach has been rendered moot by changes to Eurostat's guidance eliminating the ability to keep BBAM debt off government balance sheets via private ownership participation. For a complete discussion of the legal considerations surrounding BBAM programs in the EU, see EC 2018.

Most instances of government-owned BBAM programs involved the creation or repurposing of a special entity to acquire and manage the NPAs. A minority of programs omitted this step. Under the Uruguayan NPPS, the Central Bank of Uruguay acquired NPAs directly onto its balance sheet. In Hungary's LCPs, the Ministry of Finance acquired NPAs before transferring only a portion of them to the Hungarian Investment and Development Bank. In Mongolia, policymakers initially contemplated establishing MARA as a separately incorporated public enterprise to make it less political before deciding that it should be a government agency because they thought giving MARA governmental standing would help promote stability in the banking sector. Six years after its establishment, MARA ultimately became a public enterprise.¹⁵

Governance/Administration

A fundamental challenge of BBAM program governance is that programs must be institutionally independent given the potential for political pressure, but must also be subject to mechanisms for holding them accountable (Cas and Peresa 2016). Governance of BBAM programs must thus encompass both the internal workings of programs and their external oversight.

The use of an appropriately structured Boards of Directors to oversee BBAM programs is a key element of BBAM program governance (Ingves, Seelig, and He 2004). Important considerations include the mix between public and private sector representatives, the potential role of international experts, and the independence of directors.

¹⁴ This highlighting is not intended as an assertion that private ownership of BBAM vehicles is a promising approach. Instead, it represents the view that if private ownership is pursued, mechanisms for retaining government control over the vehicle may be an important consideration. For more details on how NAMA accomplished this control despite private ownership, please consult the underlying NAMA case study.

¹⁵ Despite representing a change to the program, this feature has not been highlighted in yellow based on our determination that the change occurred due to evolving circumstances rather than a belief by policymakers that the original design was ineffective.

Table 6 below summarizes the approaches to Board composition taken by programs studied.

Table 6: Board Composition

BBAM Program	Board Size	Details on Board Composition
Burkina Faso: BRCB	—	Not disclosed.
China: Four AMCs	—	The Ministry of Finance controlled the chairmanship of the boards of the 4 AMCs.
Colombia: CISA	5	The board had 3 public sector members from Fogafin and 2 private-sector members, with the acting president being the Minister of Finance and Public Credit.
Czechoslovakia: KOB	—	Not disclosed.
Finland: Arsenal	6	The board had 4 public-sector and 2 private-sector members. The public representatives were from the Treasury, Ministry of Finance, Ministry of Trade and Industry, and the Guarantee Fund, while 1 private-sector representative was from a bank and the other from a food/beverage packaging company.
Ghana: NPART	9	The Board of Trustees included a chairman, representatives from the Ministry of Finance and Economic Planning and the Bank of Ghana, the Chief Administrator of the Trust, a chartered accountant from the private sector, a lawyer with expertise in corporate law, and 3 other experts.
Hungary: LCPs	—	None.
Hungary: MARK	3	The board was comprised of external members of the Hungarian central bank's Monetary Council.
Indonesia: IBRA	—	A senior Ministry of Finance official served as the IBRA head, and the Financial Sector Policy Committee (FSPC) and the State-owned Enterprises Minister oversaw IBRA. The FSPC was chaired by the Coordinating Minister of Economy, and its members included the Ministers of Finance and Trade and Industry and the state Ministers of National Planning and State-owned Enterprises. IBRA also had an oversight committee, which consisted of 9 members, including a former Minister of Finance, the chairman of IBRA, and representatives from the private sector and the academic world.
Ireland: NAMA	9	The board was comprised by the CEO of NAMA, the CEO of the National Treasury Management Agency, and 7 officials appointed by the Minister for Finance.
Jamaica: FINSAC	10	FINSAC's Board of Directors was appointed by the Minister of Finance. Its original board consisted of the BOJ Governor, the BOJ Deputy Governor in charge of supervision, the managing directors of FINSAC and FIS, and some political appointees from law, labor, and academia.
Japan: RCC	—	Not disclosed.
Kazakhstan: RB	—	The Supervisory Board was chaired by a policymaker of at least "Deputy Prime Minister rank" and included members of the Economy and Finance Ministries, the central bank, the State Property Committee, and other "independent" finance and business management experts.
Kyrgyz Republic: DEBRA DRU	—	The director reported to the central bank (NBK); the Financial Sector Adjustment Credit (FINSAC) committee approved assets slated for transfer to DEBRA and oversaw broader financial sector restructuring efforts. This committee was chaired by the Prime Minister or his designee and was comprised of DEBRA's director, the Chairmen of the NBK and the State Property Fund, and the Ministers of Finance, Economy, Agriculture and Justice.
Malaysia: Danaharta	9	The board consisted of representatives from both the public and private sector, which included a chairman, a managing director (for example, the

		CEO), 2 federal government officials, 3 individuals from the private sector, and 2 international members.
Mexico: FOBAPROA	9	Four members were from the Ministry of Finance. Three were from the Bank of Mexico. Two were from the CNBV. The chairman was from the Ministry of Finance and had the deciding vote.
Mongolia: MARA	—	Not disclosed.
Nigeria: AMCON	10	AMCON's Board of Directors, which was approved by the Senate, consisted of 4 executive directors and 6 non-executive members. The central bank nominated all 4 executive directors, including the CEO, and 2 non-executive members. The Ministry of Finance nominated 3 non-executive members, including the chairman of the board, while the last nominee was nominated by the Nigeria Deposit Insurance Company.
Senegal; SNR	—	The board included representatives from the Finance Ministry, the Justice Ministry, the BCEAO, international organizations such as the World Bank or USAID, and the chair of the debt recovery committee.
Slovenia: BAMC/DUTB	7	There were 4 non-executive directors, appointed by the government, and 3 executive directors, appointed by the non-executive director.
South Korea: KAMCO	11	Members of the Management Supervisory Committee included the managing director of KAMCO; representatives from the MOFE, the Ministry of Planning and Budgeting, the FSC, and the Korea Deposit Insurance Corporation; the deputy governor of the KDB; 2 representatives from the banking industry nominated by the chairman of the Korea Federation of Banks; 3 professionals recommended by the managing director, including an attorney-at-law, a certified public accountant or a certified tax accountant, and a university professor or doctorate holder who works for a research institute.
Spain: DGF	8	Four directors were from the Bank of Spain, and the Deputy Governor was the chair. The other 4 directors were from the banking industry.
Spain: SAREB	15	The board included the chairman, the CEO, 5 independent members, and 8 "proprietary directors" who represent the main shareholders of the company.
Sweden: Securum/ Retriva	6	The chairman was a CEO of a state-owned company; other board members included the CEO of Securum, a Ministry of Finance representative, and 3 independent private-sector representatives.
Tanzania: LART	5–7	The President of Tanzania appointed the chairman of this board, and the Minister of Finance appointed between 4 and 6 other board members.
Thailand: TAMC	11	The Minister of Finance appointed up to 11 board members, including a representative from the Thai Bankers' Association, the Chamber of Commerce, and the Federation of Thai Industries.
United Kingdom: UKAR	9	UKAR shared a Board of Directors with Northern Rock Asset Management (NRAM) and Bradford and Bingley (B&B).
United States: RTC	5	Initially, the RTC was established under the oversight of the FDIC with a different board of directors. In 1991, the RTC Oversight Board was established, which included the Secretary of the Treasury as chairman; the Chairman of the Federal Reserve Board; the Secretary of Housing and Urban Development; and two private-sector representatives appointed by the President.
Uruguay: CBU	—	Not disclosed.

Source: Author analysis.

Certain programs studied were also subject to external oversight bodies, but with mixed results. In the United States, for example, the RTC had to appear before Congress semiannually to answer questions about its activities, while also submitting audited financial statements and regular reports. Congress also mandated that the RTC submit to analysis by

the Government Accounting Office (GAO). Early criticism from the GAO resulted in substantial changes to certain aspects of the RTC's operations over time, with the GAO ultimately praising the RTC for having made significant improvements in response to its oversight (GAO 1995).

In Jamaica, by contrast, a Commission of Enquiry established by Parliament to oversee FINSAC's debt recoveries failed to even produce a report and was ultimately denied additional funding.

Among the most troubled of the BBAM programs studied from a governance standpoint was Indonesia's IBRA. Despite being nominally independent, IBRA faced repeated instances of interference including the termination of its head by President Suharto. IBRA's activities were also marred by allegations of widespread corruption, resulting in lawsuits and even criminal charges.

Size

The programs studied did not generally adopt specific limitations on the amount of NPAs that could be acquired, although often they were established with some expectation for what the size would ultimately be. Those programs that did specifically restrict size did so indirectly, through caps on the amount of debt that could be issued in support of the programs' operations. Like many BBAM programs, Ireland's NAMA acquired NPAs in exchange for debt securities, but it was limited to issuing up to €54 billion in such securities before needing to seek further authorization from the government. Slovenia's BAMC similarly acquired NPAs in exchange for debt securities, with a €4 billion cap on such issuances.

BBAM programs must generally seek the middle-ground approach of being neither too small nor too large given their specific contexts. A program that is too small may fail to adequately address the banking system's NPA problem or to justify the effort and expense of a centralized solution. A program that is too large imposes a number of challenges, including increased operational difficulties and heightened risk to the taxpayer for programs that are publicly funded. As further discussed in the Eligible Assets section, although Malaysia's Danaharta did not set an explicit program size, it determined an amount of NPAs that it thought was manageable and then established asset size thresholds based on an analysis of how many NPAs it would end up acquiring based on those thresholds.

Funding Source

Consistent with the fact that the majority of BBAM programs studied were fully publicly owned, most of the programs studied were also fully publicly funded. This public funding could come directly from the taxpayer, be raised with the issuance of government bonds, take the form of government-guaranteed bonds issued to the transferring institutions as consideration, or be paid for by the central bank (sometimes with the government later taking on the central bank's obligations). However, even some publicly owned programs incorporated features intended to share losses with private parties. Funding programs through assessments on participating banks or the financial system more broadly might

provide important political and communication leverage because those viewed as responsible for the problem are seen as having to pay for addressing it, but this must be weighed against the risks associated with placing an additional financial burden on the very institutions that are weak and in need of rescue.

Ireland's NAMA, in addition to being majority privately owned, included several mechanisms for sharing losses. If NAMA ended up sustaining losses in the aggregate, it had the ability to tax participants to be made whole. The consideration NAMA paid for acquired assets was also 5% comprised of subordinated debt issued by NAMA and tied to its financial performance (with the remaining 95% in government-guaranteed bonds). Several programs sought to relieve the financial burden on the government over time through broad, ongoing assessments on the financial sector. Nigeria established a Banking Sector Resolution Cost Fund (RCF) alongside publicly owned AMCON to absorb the losses on acquired NPAs. The RCF was funded over a 10-year period with contributions from the Central Bank of Nigeria (33% of RCF funding) and from annual 30-to-50-basis-point assessments on the assets of all institutions eligible for AMCON (67% of RCF funding). Mexico funded publicly owned FOBAPROA with 30-to-70-basis-point assessments on the liabilities of banks. Spain funded the publicly owned Deposit Guarantee Fund with equal contributions from the Bank of Spain and the Spanish banking sector, with banks subject to a 10-to-30-basis-point assessment on deposits. Such efforts were not always enough to prevent further losses to the taxpayer. The RCF, for example, proved insufficient to ultimately cover all of the losses associated with Nigeria's combined BBAM and recapitalization initiative.

And as will be discussed in more detail in the section on Pricing, multiple programs studied contained ex post purchase price adjustments intended to minimize losses in the first place.

Inadequate and/or uncertain funding can undermine BBAM programs by interfering with their ability to conduct operations and reducing their willingness to dispose of assets that will impose additional losses. The need to go back for additional funding may also increase the risk of a loss of political independence. Analysts have cited funding problems as undermining effectiveness in evaluations of programs including the United States' RTC (FDIC 1998) and Mexico's FOBAPROA (Klingebiel 2000). Hungary's proposal to temporarily fund asset purchases with bridge loans from the Hungarian central bank violated the European Central Bank's prohibition on monetary financing and prevented Hungary's MARK from conducting operations.

Eligible Institutions

As illustrated in Table 1 in the Overview, the programs studied include a mix of open-bank programs, closed-bank programs, and programs for both open banks and closed banks. This determination would typically follow from a diagnosis of the situation through a stress test, asset quality review, or some other method. Having ascertained if there are banks needing to be liquidated, banks needing support, or some combination of both, policymakers can determine if a BBAM program is appropriate and, if so, define eligibility accordingly.

Programs were generally available to banks in the relevant jurisdiction (or to failed banks in the case of closed-bank programs), with some exceptions. Even some open-bank programs limited participation to institutions that met certain criteria for distress, including falling below required capital adequacy ratios (as was the case with the bank-oriented component of Hungary's LCPs) or failing stress tests (as was the case with Spain's SAREB). Other open-bank programs, such as BAMC in Slovenia and China's 1999 AMC program, targeted the largest financial institutions in the country.

Whether or not participating institutions are state-owned can be an important consideration, with implications for other design features, such as the valuation of the acquired assets as discussed in Pricing. Given the widespread use of BBAM programs in countries transitioning away from state-controlled financial systems, several programs studied involved significant participation by state-owned banks. In Tanzania, LART initially limited participation to state-owned institutions, before expanding eligibility to private institutions after the program was underway.¹⁶

Nonbank financial institutions were included in a minority of programs studied. South Korea's KAMCO allowed nonbank financial institutions, including investment trust companies, insurance companies, and securities firms, to participate, although it ended up acquiring most assets from banks. The definition of eligible financial institutions under Japan's Resolution and Collection Corporation (RCC) encompassed certain specialized institutions, such as agricultural and fishery cooperatives. Thailand's TAMC included state-owned and private AMCs because the country initially attempted a decentralized approach to dealing with its NPA problem before deciding that a centralized AMC was necessary.

A key consideration in defining eligible institutions is whether to make participation in a BBAM program mandatory or voluntary. For the programs studied that involved a closed-bank component, participation was generally automatic upon an institution's reaching a given state of failure. For open-bank programs, participation was sometimes mandatory for institutions meeting certain requirements.

For voluntary programs, the price paid for acquired NPAs is obviously a primary driver of participation as is discussed in Pricing. Among the programs studied, policymakers sometimes also included incentives for participation. One common incentive was requiring banks seeking to take advantage of the concurrent recapitalization program to participate in the BBAM program. This was the case for Malaysia's Danaharta, which also had a multi-faceted "carrot and stick" approach to encourage eligible institutions to participate. In addition to a profit-sharing system discussed in more detail in Pricing, Danaharta allowed institutions transferring NPAs to amortize the losses over a five-year period. Conversely, any NPAs not transferred had to be written down immediately. Institutions also had only one opportunity to transfer a given NPA. Additional tools used in other programs studied include

¹⁶ Despite representing a change to the program, this feature has not been highlighted in yellow based on our determination that the change occurred due to evolving circumstances rather than a belief by policymakers that the original design was ineffective.

capital surcharges on targeted assets remaining on bank balance sheets (Hungary's MARK) and caps on NPA levels (Nigeria's AMCON).

Some programs approached eligibility by targeting debtors rather than creditors. In addition to a bank-oriented component that took the traditional approach of acquiring NPAs from targeted banks, Hungary's LCPs had a firm-oriented component that acquired NPAs made to select "strategic" state-owned enterprises regardless of the originating bank. Kazakhstan's Rehabilitation Bank focused on the country's largest delinquent debtors.

Eligible Assets

As noted in the Background section, the types of eligible assets can determine the success of a program, with certain types of assets being particularly ill advised. The assets eligible to be included must thus be defined with particular care. **Among the asset types that may be a poor fit for BBAM programs are:**

- Loans to state-owned enterprises or other politically connected debtors, which can increase the likelihood of government interference, as was the case in several programs studied including IBRA in Indonesia, Société Nationale de Recouvrement (SNR) in Senegal, and FOBAPROA in Mexico.
- Residential mortgages, which can also be politically sensitive, as was the case in Jamaica's FINSAC.
- Consumer loans and loans to small- and medium-sized enterprises, which may require too much direct knowledge of the debtors.

Assets typically better suited to BBAM programs include commercial real estate and large corporate exposures, which are generally larger in size, less politically fraught, and backed by collateral that is easier to realize upon.

Additional complications can arise if the assets acquired are too heterogeneous or include too many small exposures to be effectively managed. Finland's Arsenal accepted NPAs of all types and sizes, a fact that evaluations of the program have cited as hindering its efforts to dispose of assets (Klingebiel 2000). To address the concern about the efficiency of handling small assets, several programs studied included minimum asset size requirements including Securum/Retriva in Sweden, Danaharta in Malaysia, and SAREB in Spain. In Malaysia, policymakers established the minimum size threshold at the point at which they thought they could manage the workload, having analyzed what loans that meant they would be buying.

Certain programs studied went beyond simply excluding small exposures and specifically targeted NPAs from the most significant delinquent debtors in the country. This was the approach taken by Kazakhstan's Rehabilitation Bank and the firm-oriented component of Hungary's LCPs. The ability to aggregate the debt of such large debtors and thereby increase negotiating power is one potential benefit of a BBAM approach.

One of the potential concerns associated with BBAM programs is the risk of moral hazard. If banks have an expectation that they will be able to transfer NPAs to a BBAM program, they may be more likely to extend risky loans. Several programs studied sought to address moral hazard by limiting eligibility to assets initiated by a given date. China's 1999 AMC program accepted only NPAs generated by policy lending prior to 1996. The bank-oriented component of Hungary's 1992 LCPs excluded loans made after October 1, 1992. Tanzania's LART initially restricted eligibility to assets in existence at the time of the legislation establishing the program.

The risk associated with being too restrictive in the assets accepted by a BBAM program is that it may leave banks with NPA levels that remain problematically high. This appears to have been the case with Thailand's TAMC, in which different sets of asset eligibility criteria were developed for state-owned institutions and for private institutions. The criteria for private institutions were more stringent, with NPAs subject to a previous court ruling or restructuring agreements deemed ineligible. Given that Thailand had already been in crisis for several years by the time of the TAMC's introduction, this excluded a majority of NPAs held by private institutions. As a result, the TAMC acquired the vast majority of its assets from state-owned institutions, while private banks and AMCs retained high levels of NPAs.

While the nature of BBAM programs is such that they typically acquired NPAs specifically, many programs, such as the RTC in the United States, the RCC in Japan, and the BAMC in Slovenia, also acquired performing loans. This may have been motivated by the belief that consolidating all of the obligations of a given debtor, whether performing or not, would give the BBAM program additional leverage and promote more effective restructuring. This was the motivation explicitly cited by the BAMC. It may also be intended to help the BBAM program achieve stand-alone profitability. However, there may be circumstances in which specific performing assets are included for political reasons. Nigeria's AMCON purchased performing loans that it deemed too large relative to the capital base of the transferring banks, but critics allege that these assets were included because a government official was the debtor (Cerutti and Neyens 2016).

Acquisition Mechanics

Policymakers designing BBAM programs must determine the mechanics of acquiring the targeted assets. The approaches here are generally very different for closed-bank programs and open-bank programs. Closed-bank programs typically acquire assets automatically upon an institution's failure, often with no need to make decisions about which specific assets to acquire and what consideration to provide. In some instances, negotiation over which NPAs a closed-bank program would take on were a key feature of attempts to sell failed banks to acquiring institutions, as was the case for the United States' RTC and the Spain's Deposit Guarantee Fund.

Open-bank programs, conversely, must choose which NPAs to acquire, as well as the timing, logistics, and consideration to be used. In many cases, participating institutions provided BBAM programs with lists of eligible assets that the programs would then evaluate for acquisition. In Hungary, MARK selected the specific assets to be acquired at random off a list

of all eligible assets to avoid adverse selection and help ensure that assets were acquired at an appropriate price.¹⁷

The timing of asset acquisition by open-bank BBAM programs can create moral hazard concerns. Ongoing asset purchases may encourage banks to engage in risky lending with the expectation that any NPAs that result can be transferred to the program. For programs that want or need to make ongoing purchases, this can be addressed by limiting the assets that can be transferred to those existing as of a certain date, as discussed in Eligible Assets. Several programs did in fact employ a tranche approach, in which assets were purchased in groups over time, in most cases seemingly to avoid the logistical difficulties that would have been associated with attempting to acquire all targeted NPAs at once. Malaysia's Danaharta acquired NPAs in two rounds—an initial primary carveout and a secondary carveout with more selective criteria. Thailand's TAMC sorted NPAs into tranches based on the number of creditors, institution type, and amount of outstanding debt. Slovenia's BAMC divided purchases among multiple tranches, with all of the assets associated with a given debtor included in the same tranche. Spain used the health of the transferring financial institutions to split SAREB's acquisition into two groups, with the most distressed institutions transferring first. In Ireland, NAMA had to abandon its initial plan of sorting assets into tranches where the first tranche contained a small number of debtors with large exposures and later tranches would include more debtors with smaller exposures. This multi-step process was proving to be too slow. The need to accelerate the pace of asset acquisition resulted in the collapsing of multiple anticipated future tranches into one final bulk tranche.

The consideration paid for acquired NPAs in open-bank programs typically consisted of debt securities, often guaranteed by the government. The use of debt securities avoids the need to provide cash up front and instead allows for payment over time as the program begins to realize returns on acquired NPAs. This is a concern primarily for budget-constrained governments that would otherwise not have the ability to fund the purchases. In such situations, bond maturity should be tied to the lifespan of the program and the expected timing of cash flows from the NPAs (Ingves, Seelig, and He 2004). Another approach is to use zero-coupon bonds to delay the need to make any payments until bond maturity, as was done in Nigeria's AMCON. If the bonds are eligible collateral for central bank lending facilities, they can also be used to improve recipient banks' liquidity position right away. A less common form of consideration seen in Kazakhstan's Rehabilitation Bank and Senegal's SNR was the removal of NPAs from bank balance sheets in exchange for the cancellation of a corresponding amount of bank liabilities owed to the government. These liabilities could

¹⁷ While the random selection of assets off a submitted list to reduce adverse selection was a key feature highlighted by the European Commission in its consideration of Hungary's MARK, the ability of random selection to effectively combat adverse selection is dependent on the submitted list itself not being the product of adverse selection. Randomly selecting assets off a list of an institution's worst NPAs would still leave a BBAM program with the worst NPAs. Such a design feature is thus interesting, but also to be approached with caution.

take the form of previous government assistance and/or deposits from state-owned enterprises or other agencies of the government.¹⁸

Acquisition Pricing

A key consideration for the design of a BBAM program, particularly one that will engage in open-bank acquisitions, is the price at which assets will be acquired. Cerruti and Neyens (2016) have described this as the most difficult of all of the design decisions associated with establishing a BBAM program. As discussed throughout this survey, it is also a decision that has significant implications for other program features.

As illustrated in Table 7, approaches to pricing can generally be divided into one of three categories: acquisition at book value, acquisition at real economic value, or acquisition at market price. In crisis conditions, it is generally understood that acquisitions at both book value and market price are likely to result in pricing that is different from the actual value of the asset. If an NPA is acquired at book value, the price will be higher than the asset's ultimate return, given its nonperforming status. This shifts the losses associated with the NPA from the transferring institutions to the BBAM program. If an NPA is acquired at market price, the price will be lower than the asset's ultimate return, given the market-distorting effect of the crisis.

¹⁸ *The ability to net the purchase price for transferred assets against amounts already owed by the transferring institutions to the state may be a way of simplifying the process of acquiring assets in some circumstances.*

Table 7: Approaches to Pricing

Book Value	Real Economic Value	Market Price	Unique Method	Unknown/Unclear
China: Four AMCs (Initial) ^a Finland: Arsenal Ghana: NPART Jamaica: FINSAC Mexico: FOBAPROA ^b Senegal: SNR Spain: DGF Sweden: Securum/Retriva	Ireland: NAMA Slovenia: BAMC Spain: SAREB	China: Four AMCs (Subsequent) ^a Hungary: MARK Japan: RCC (Subsequent) ^c South Korea: KAMCO ^d Malaysia: Danaharta Nigeria: AMCON Thailand: TAMC	Hungary: LCPs ^e Indonesia: IBRA ^f	Burkina Faso: BRCB Colombia: CISA Czechoslovakia: KOB Japan: RCC (Initial) ^c Kazakhstan: RB Kyrgyz Republic: DEBRA Mongolia: MARA Tanzania: LART United Kingdom: UKAR Uruguay: CBU ^g United States: RTC ^g

Notes:

- China's AMCs used book value for the first wave of asset purchases and an auction-based method for the second wave of acquisitions.
- FOBAPROA acquired assets at net book value, calculated as the book value less loan-loss provisions.
- Prior to 2002, the RCC purchased assets at below market value but an amendment in 2002 allowed it to purchase assets at market price.
- KAMCO sorted assets into different categories based on the asset type: ordinary loans, restructured loans, out-of-court workout loans, and those connected to a large conglomerate. The purchase price depended on when the loan was acquired (the earlier or later tranches) and whether or not it was secured.
- Under the LCPs in Hungary, the government purchased assets with a haircut on book value depending on when the debt was classified as bad. Some debt was purchased at 100% of book value.
- Institutions in Indonesia transferred assets to IBRA at zero value but would receive proceeds from the ultimate disposal.
- The RTC and CBU dealt exclusively with assets from failed banks, and the pricing methods appear to be less clear because of the nature of the intervened banks. The pricing method is also less important for agencies dealing with only closed-bank assets.

Source: Author analysis.

Both paying too much and paying too little for NPAs have the potential to undermine the effectiveness of BBAM programs. Paying too much may obviate the need for recapitalizing the transferring banks as discussed in the Part of a Package section, but there are potential concerns associated with this form of indirect recapitalization. These include (1) the fact that

capital provided in this form is free or low cost and therefore disadvantages other firms and (2) the fact that direct injections typically afford policymakers more control over the institutions receiving the capital (Rose 2005). Paying too much may also undermine attempts to deal with the NPAs themselves. If BBAM programs acquire NPAs at prices above what can be realized from the assets, the disposal of the NPAs will result in large, immediate losses. The desire to avoid such an outcome may result in BBAM programs' holding on to NPAs without any real effort to dispose of them. If acquiring NPAs at higher values is necessary despite this concern, an immediate revaluation on the books of the BBAM program may be important in removing any disincentive to disposal (Rose 2005).

As indicated in Table 7, acquisition at book value has been a frequently used approach, notwithstanding the concerns associated with paying too much for NPAs. Sometimes this may be a practical consideration. The Spanish Deposit Guarantee Fund, for example, purchased NPAs at book value because policymakers considered it too difficult to determine an alternate, market-based price. It is also the case that book value pricing may be less of a concern where the participants in a BBAM program are state-owned banks, because the state will be responsible for the losses whether they are imposed on the program or on the transferring banks.

Paying too little for NPAs introduces its own set of potential challenges. If institutions perceive the price offered for NPAs as too low and participation is voluntary, there may be little uptake. This was initially the case for the RCC in Japan, which in its early years paid an average of 3.8% of book value and attracted few participants. Japanese policymakers later successfully amended the RCC's approach with the explicit objective of encouraging more participation. Prices can be too low to secure participation not only in absolute terms, but also relative to institutions' expectations about what prices will be available in future versions of the BBAM program. Hungary's LCPs featured both an initial bank-oriented component that acquired NPAs from targeted banks and a subsequent firm-oriented component that acquired NPAs made to targeted state-owned enterprises by any banks. It appears that banks were reluctant to dispose of eligible NPAs during the initial bank-oriented component because they expected those same NPAs to fetch higher prices during the subsequent firm-oriented component.

Even if BBAM programs succeed in attracting participants at low prices (possibly because participation is mandatory), such pricing can interfere with the effective disposition of acquired NPAs. A BBAM program that acquires NPAs at very low prices may be able to achieve significant profits without maximizing the value of those assets. This could reduce incentives for maximizing value and promote hurried disposal at prices below those that might ultimately be possible.

The real economic value approach to pricing can be seen as an attempt to find the appropriate middle path between prices that are too high and prices that are too low for a BBAM program to be successful. Typically it involves an attempt to determine the intrinsic value of an asset outside of crisis conditions when the market-distorting effects of the crisis have subsided and the actual losses incurred by the asset are realized. This can prove difficult, as was the case with Spain's SAREB, where a revaluation of assets three years after

they had been acquired at what was then considered real economic value resulted in substantial losses.

For BBAM programs introduced in the European Union, purchasing assets at an above-market price constituted State aid under EC rules. Such purchases would be approved only if they met certain criteria set forth in the Communication from the Commission on the Treatment of Impaired Assets in the Community Banking Sector of February 25, 2009 (better known as the Impaired Asset Communication, or IAC) and subsequently modified in future communications. To comply with these criteria, programs generally needed to purchase assets at their “real economic value,” a figure aiming to approximate the asset’s long-term economic value based on prudent assumptions about cash flows and observable market inputs. This resulted in lengthy valuation processes and/or clashes with the European Commission that often delayed BBAM programs in the EU. For a more detailed discussion of the European Commission’s approach to pricing, see EC 2018.

Given the uncertainty and complexity associated with attempting to value assets in the midst of crisis and often with little time to do so, some BBAM programs relied on ex post purchase price adjustments based on how the NPAs ultimately performed. These could be structured to protect the BBAM program from losses resulting from overpaying, to protect the transferring banks from losses resulting from underpaying, or both. Sometimes these ex post adjustments were done only temporarily given the need to acquire an initial set of assets with considerable speed. This was the case with South Korea’s KAMCO, which aimed to make rapid initial bulk purchases, leaving inadequate time to completely evaluate assets. The AMC consequently included mechanisms for ex post price adjustments. Once market conditions stabilized, KAMCO shifted its approach to rely on more exact upfront pricing with no adjustments.¹⁹

Other adjustments were more permanent. Perhaps the most extreme adjustment example was Indonesia’s IBRA, under which banks received no upfront consideration for the transferred NPAs but were entitled to any proceeds from their ultimate disposition. In Ireland, NAMA set an initial price for assets based on calculations of real economic value, but this amount was subject to a clawback feature that enabled NAMA to recover any amount it was determined to have overpaid on acquired assets stemming from expedited and/or inaccurate initial valuations. NAMA could also include a clawback feature to protect banks from underpayment, but it is unclear how widely used this was. Danaharta in Malaysia included a profit-sharing arrangement as an incentive for participation, whereby transferring institutions were entitled to 80% of the profits on NPA disposal. Conversely, Mexico’s FOBAPROA negotiated loss-sharing arrangements with participating banks that would leave banks responsible for 20 to 30% of the losses on the NPAs that they transferred.

¹⁹ Despite representing a change to the program, this feature has not been highlighted in yellow based on our determination that the change occurred due to evolving circumstances rather than a belief by policymakers that the original design was ineffective. In a situation in which initial large-scale asset purchases must be done quickly without adequate time for valuation, it may be worth considering making the use of purchase price adjustments temporary and transitioning to full valuation with no adjustments as soon as conditions allow.

Under Thailand's TAMC, transferring institutions shared in profits up to the book value of transferred assets and were responsible for losses of up to 30% of the transferring price. The TAMC and the transferring institutions calculated and recognized these profits and losses at the end of the fifth and 10th years of the TAMC's operations.

While potentially attractive in terms of their ability to promote accurate pricing and protect BBAM programs from losses, adjustments may hinder the effective functioning of the intervention. Banks may be less willing to transfer NPAs if they remain exposed to further losses on them (Ingves, Seelig, and He 2004). And even if banks do transfer the NPAs, this continued exposure could mean that counterparties remain uncertain about the banks' viability. Particularly where a BBAM program is intended to restore counterparty confidence in institutions to, for example, ensure their continued access to funding, ex post price adjustments may be counterproductive.

Management and Disposal

The approaches taken to the management and disposal of acquired NPAs in programs studied were closely related to their overarching mandates. An overview of select programs' approaches is included in Table 8.

Table 8: Approaches to Management and Disposal

Management or Disposal Tools	Select Examples and Descriptions
Auctions	<u>Indonesia: IBRA</u> IBRA sold corporate loans and some SME loans through open auctions. <u>South Korea: KAMCO</u> KAMCO sought to foster a market for distressed assets and held auctions to promote international and domestic participation.
Bulk Sales	<u>Mongolia: MARA</u> MARA intended to establish a bulk sales program to dispose of NPAs it acquired. <u>Spain: SAREB</u> SAREB initially packaged assets in large portfolios for fast sales to service the AMC's debt burden.
Contracting or Outsourcing for Disposal and Management	<u>Hungary: LCPs</u> For the majority of NPAs transferred, the Ministry of Finance entered into temporary management contracts with the transferring financial institution. <u>US: RTC</u> The RTC entered into three-year arrangements with private contractors who managed pools of nonperforming assets.
Debt Collection	<u>Nigeria: AMCON</u> AMCON focused on debt collection because a significant amount of AMCON's portfolio could be traced to a small number of borrowers.
Foreclosures and Seizing Assets	<u>Burkina Faso: BRCB</u> The BRCB had special authority to seize assets to recover the value of assets from delinquent borrowers.
Initial Public Offerings	<u>Sweden: Securum/Retriva</u> Securum sold a total of five of its subsidiary real estate management companies on public exchanges (four in Sweden and one in London).
Negotiated Private Sales	<u>Ghana: NPART</u> NPART used a negotiated sales method for a majority of the sales it conducted.
Real Estate Management	<u>Sweden: Securum/Retriva</u> Securum/Retriva used subsidiary real estate management companies for properties in different regions or related to specific industries.
Restricted Tender Exercises	<u>Malaysia: Danaharta</u> Danaharta disposed of foreign loans and marketable securities through restricted tender exercises.
Restructuring/ Workout	<u>Ireland: NAMA</u> Borrowers submitted business plans to NAMA, which evaluated the plans and determined if full or partial restructuring was viable. <u>Thailand: TAMC</u> The TAMC had broad authority to restructure debt, including reducing principal or interest, extending repayment periods, making debt-to-equity conversions and lease-back arrangements, among others.
Securitization	<u>Malaysia: Danaharta</u> To expedite the recovery of restructured loans, Danaharta transferred a portfolio of such loans to an SPV that issued the first collateralized loan obligation in Malaysia. <u>US: RTC</u> The RTC used securitization to dispose of a variety of assets and eventually issued portfolio performance reports that became industry standards for issuers of securities.

Source: Author analysis.

A minority approach to the management and disposal of NPAs gave responsibility for the assets to the transferring institutions, either because the institutions were better positioned to deal with them and/or because the exposures were too small to warrant the attention of BBAM program staff. Where a primary objective of a BBAM program is to free transferring banks to focus on their core business, requiring these banks to remain responsible for the management and disposition of the assets can be self-defeating. Such arrangements also require careful consideration of the banks' incentives and contractual provisions that reflect these realities. Hungary's LCPs adopted this approach for the bank-oriented component of its program. Mexico's FOBAPROA also made use of the banks themselves, and as discussed in the Pricing section, negotiated loss-sharing arrangements to incentivize banks to manage assets appropriately. In Spain, SAREB initially relied on transferring banks for management and disposal before concerns about conflicts of interest ultimately prompted a shift to outside contractors.

The use of outside contractors or the sale of NPAs to third parties who will then seek to collect on them can create political challenges. In Jamaica, FINSAC transferred residual NPAs to an American financial institution whose aggressive collection efforts resulted in widespread criticism from the Jamaican public.

In some instances, attempts to build safeguards into the disposal process interfered with program effectiveness. Congress initially required the United States' RTC to dispose of assets for at least 95% of assessed value. They later reduced this threshold to 70% to accelerate the disposition process. In Slovenia, the BAMC was subject to a requirement that it dispose of at least 10% of its assets every year, but this may have resulted in forced sales and unnecessary losses.

The need to dispose of large volumes of NPAs often resulted in the pioneering of new techniques and products. The United States' RTC is often credited with having jumpstarted the market for commercial mortgage securitizations. Danaharta was responsible for the first collateralized loan obligation in Malaysia. The desire to develop secondary markets for NPAs and new approaches for dealing with them was often a key consideration in launching BBAM programs.

Timeframe

A final key consideration in the design of BBAM programs is whether or not the program should have a predefined timeframe and, if so, how long it should be. Too short a timeframe could result in a program's disposing of NPAs prematurely in ways that exacerbate market conditions and/or fail to maximize the value of the assets. Too long a timeframe could result in NPAs' languishing in the program and continuing to lose value over time.

As Table 9 illustrates, the programs studied took a variety of approaches to the timeframe question. Given the potential concern that too long a timeframe could result in languishing, it is perhaps surprising to see how many programs had no predefined timeframes and were effectively open ended. The IMF repeatedly criticized Nigeria's AMCON, for example, for failure to establish a sunset date, warning that it increased financial risks and the threat of moral hazard (IMF 2013). Instead of relying on open-ended timeframes, several programs

studied appear to have addressed timing uncertainty by establishing a defined lifespan subject to extension. As indicated in Table 9, several countries made use of such extensions, while others ended programs early.

Table 9: Approaches to Timeframe

BBAM Program	Predefined Timeframe (Approximate)	Actual Timeframe (Approximate)
Burkina Faso: BRCB	None	11 years
China: Four AMCs	10 years	6.5 years
Colombia: CISA	None	7 years
Czechoslovakia: KOB	4 months	2 years
Finland: Arsenal	Estimated 5-7 years	10 years
Ghana: NPART	5 years	6.5 years
Hungary: LCPs	None	Unknown
Hungary: MARK	10 years	Never operated
Indonesia: IBRA	5 years	6 years
Ireland: NAMA	None	Ongoing as of 2021
Jamaica: FINSAC	Estimated 5-7 years	5 years (ongoing litigation for additional 14 years)
Japan: RCC	7 years	Unknown
Kazakhstan: RB	4 years	Unknown
Kyrgyz Republic: DEBRA DRU	3 years	Unknown
Malaysia: Danaharta	None	6.5 years
Mexico: FOBAPROA	None	4 years
Mongolia: MARA	3 years	Unknown (at least 9 years)
Nigeria: AMCON	None	Ongoing as of 2021
Senegal: SNR	None	Ongoing as of 2017
Slovenia: BAMC/DUTB	5 years	Ongoing as of 2021
South Korea: KAMCO	None	16 years
Spain: DGF	None	Unknown
Spain: SAREB	Estimated 15 years	Ongoing as of 2021
Sweden: Securum/ Retriva	Estimated 10-15 years	5 years
Tanzania: LART	4 years	10 years
Thailand: TAMC	12 years	12 years
United Kingdom: UKAR	None	11 years
United States: RTC	6.5 years	5.5 years
Uruguay: CBU	None	Unknown

Source: Author analysis.

III. Evaluation

A small body of literature has examined BBAM programs to determine which are generally seen as successful, which are generally seen as unsuccessful, and what shared characteristics may account for their relative performance. A 2016 World Bank study (Cerutti and Neyens) compared nine centralized AMCs from the 1980s through the Global Financial Crisis and, as

outlined in Table 10, identified non-exhaustive “good practices” for AMC management. A 2020 IMF report (Dobler, Moretti, and Piris) provided lessons learned from the organization’s official response to systemic banking crises across the world and came up with a similar list of design features of successful centralized AMCs. A 2017 Financial Stability Institute report (Baudino and Yun) described various options for addressing systemic NPA problems and the situations in which centralized AMCs might be necessary and most likely to succeed. In 2018, based on the recent experience of Ireland and other EU countries with centralized AMCs, the European Commission published an “AMC Blueprint” with wide-ranging recommendations for EU countries considering establishing AMCs in the future (EC).

These studies identified many of the same context factors and design decisions that our survey has shown to be particularly influential in determining program success. Programs whose NPAs involve commercial real estate or large corporate exposures are generally more successful than programs whose NPAs comprise loans to politically connected debtors, residential mortgages, consumer loans, or SME loans. Programs in countries with strong legal systems are generally more successful than programs in countries with weak legal systems in which creditors cannot effectively enforce their rights. Programs with strong governance and good transparency are generally more successful than programs without.

Yet, despite the frequency with which they have been used, dating back to as early as the 1980s, BBAM programs have not been the subject of many empirical studies that seek to demonstrate whether or not they worked and, if so, what design features promote program success (Brei et al. 2020). This may stem in part from the challenges associated with evaluating BBAM programs. As discussed in Part of a Package, BBAM programs are often combined with other interventions such that it can be difficult to disentangle the effects of different parts of a combined package. BBAM programs are also highly context specific. Two similar programs may produce very different outcomes based on the nature and extent of the problem NPAs and the speed of the economic recovery. There is also the question of what it even means for a given BBAM program to be a success in light of the various objectives policymakers may have had in introducing it. The political and legal climate matters. A BBAM program may succeed at cleaning up balance sheets and spurring lending, but at a fiscal cost that taxpayers deem unacceptable.

So how to measure success? The World Bank study (Cerutti and Neyens 2016) noted that BBAM assessment traditionally focused on recovery rates on impaired assets, but it is now more likely to consider a program’s holistic contribution to resolving a crisis, taking into account its ability to repay its own debt and limit taxpayers’ losses. The study presented high-level statistics measuring these factors for the nine programs it examined.

In one of the few empirical studies of a much larger dataset, Brei et. al. (2020) analyzed BBAM programs’ success at cleaning up bank balance sheets and encouraging continued lending, reflecting mandates common to all BBAM programs. Brei et. al. used a dataset of 135 banks from 15 European banking systems during the period from 2000 to 2016 that benefited from “asset segregation” initiatives. Brei et. al. defined asset segregation broadly to include both centralized BBAM programs and decentralized AMCs. Brei et. al. find that asset segregation

without recapitalization has a statistically insignificant impact on future NPA levels and loan growth. Recapitalization-only approaches, in turn, improved NPA levels but not loan growth, as banks use injected funds to clean up balance sheets rather than extend new lending. Thus, neither tool is effective on its own in reducing future NPAs and promoting lending. Instead, they must be used together to achieve lower NPA levels and boost credit growth. Brei et. al. also find that asset segregation is more effective when it (1) is privately funded²⁰, (2) involves smaller shares of originating banks' assets, and (3) occurs in countries with strong legal systems.

The effective use of BBAM programs by policymakers requires a thorough analysis of the specific context they face and, if a BBAM program is indicated, the careful development of a program designed to maximize effectiveness.

²⁰ As noted, historically, most BBAM programs have been publicly funded. Given that Brei et. al. (2020) include both decentralized AMCs and BBAM programs in their study, it could be that the effectiveness of private funding is a relevant conclusion only for the former.

Table 10: Good Practices for BBAM Programs

Issue	Practice	Examples
Preconditions	Strong political consensus	Ireland: NAMA South Korea: KAMCO US: RTC
	Willingness to recognize losses	Sweden: Securum/Retriva
	Comprehensive reform program	Malaysia: Danaharta Sweden: Securum/Retriva
	Strong supervision and insolvency framework	Ireland: NAMA
	Solid diagnosis and critical mass of impaired assets	Malaysia: Danaharta Spain: SAREB
	Strong tradition of independence and accountability	Sweden: Securum/Retriva
	Robust legal framework for bank resolution, debt recovery, and creditor rights	South Korea: KAMCO US: RTC
Legal and Institutional Framework	Clearly focused and narrow mandate	Malaysia: Danaharta Sweden: Securum/Retriva
	Use of special powers is limited and subject to oversight	Malaysia: Danaharta
	Strong commercial focus, private-sector experience	Ireland: NAMA Malaysia: Danaharta Spain: SAREB
	Strong levels of governance, frequent reporting, including annual financial statements	Ireland: NAMA Malaysia: Danaharta South Korea: KAMCO Spain: SAREB
	Adequate funding for operating expenses	Sweden: Securum/Retriva
	Transfer price based on market value using independent experts	Ireland: NAMA Spain: SAREB
	Adequate safeguards	South Korea: KAMCO Spain: SAREB
Operational Considerations	Use of private contractors where possible, with oversight	Malaysia: Danaharta US: RTC
	Development of strategic plan and detailed business plans	Ireland: NAMA
	Strong internal controls and transparency	US: RTC
	Early focus on asset disposition	Malaysia: Danaharta South Korea: KAMCO Sweden: Securum/Retriva US: RTC
	Transparency and availability of AMC's history, operations, and reports (such as publishing online)	Indonesia: IBRA Malaysia: Danaharta South Korea: KAMCO US: RTC

Source: Cerruti and Neyens 2016.

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